Appendix: Conservative Event-Level Timing Audit

APPENDIX A
CONSERVATIVE EVENT-LEVEL TIMING AUDIT

Each logged REFIT row provides a common timestamp but not the physical acquisition time of every individual meter reading. Consequently, the recorded data cannot identify an exact, universal IAM–aggregate delay. A single nearby aggregate step is also insufficient evidence for a shift, since an unmonitored load can produce that step. We therefore describe the following procedure only as a conservative, diagnostic event-level audit; it is not part of the primary training or evaluation pipeline and does not claim physical synchronization.

Let x_t and $P_{t,i}$ denote the recorded aggregate and appliance powers, respectively, and define $\Delta x_t = x_t - x_{t-1}$. Consider a complete on period $[s, e)$ of appliance i, with a rising edge $a_i^{\uparrow} = P_{s,i} - P_{s-1,i}$ and a falling edge $a_i^{\downarrow} = P_{e-1,i} - P_{e,i}$. Only events whose two appliance edges are at least $\max(40, 0.3\tau_i)$ W are considered, where τ_i is the appliance on threshold. We examine lags $\ell \in \{-2, -1, 0, 1, 2\}$, corresponding to at most ± 16 s. A lag is admissible only if both aggregate edges have the expected direction and sufficient magnitude:

$$\begin{aligned} \Delta x_{s+\ell} &\geq \max(20, 0.25 a_i^{\uparrow}), \\ -\Delta x_{e+\ell} &\geq \max(20, 0.25 a_i^{\downarrow}). \end{aligned} \tag{1}$$

For every admissible lag, the two-edge discrepancy is

$$C_i(\ell) = \frac{1}{2}\left(\frac{|\Delta x_{s+\ell} - a_i^{\uparrow}|}{\max(a_i^{\uparrow}, 40)} + \frac{|-\Delta x_{e+\ell} - a_i^{\downarrow}|}{\max(a_i^{\downarrow}, 40)} \right). \tag{2}$$

The selected lag is $\ell^* = \arg\min_\ell C_i(\ell)$, with zero lag preferred only when scores are tied. We accept it only when $C_i(\ell^*) \leq 0.75$, its margin over the next-best admissible lag is at least 0.20, neither event boundary lies near a ≥ 40 W transition of another monitored target, and the shifted event would not overlap an existing on period. Events failing any condition are left unchanged.

For an accepted non-zero lag, the aggregate is immutable. Let b_i be the median measured power in the nearby off samples and let $q_{t,i} = \max(P_{t,i} - b_i, 0)$ within $[s, e)$. The procedure moves the excess event profile as a whole:

$$\widetilde{P}_{t,i} = P_{t,i} - q_{t,i}\mathbf{1}_{[s,e)}(t) + q_{t-\ell^*,i}\mathbf{1}_{[s+\ell^*,e+\ell^*)}(t), \qquad \widetilde{x}_t = x_t. \tag{3}$$

Thus $\sum_t \widetilde{P}_{t,i} = \sum_t P_{t,i}$ for every shifted event. The associated binary on period is shifted by the same lag. This preserves the recorded input and the appliance watt-sample energy; it does *not* impose $x_t \geq \sum_i P_{t,i}$, because the measured target set is incomplete and the recorded aggregate must not be reconstructed using its labels.

The audit is a sensitivity analysis, not part of the retained training pipeline. In our cross-house ablation it slightly changed local timing errors but did not consistently improve F1 or AP across the two held-out datasets. The primary results therefore use the unshifted, mean-resampled recordings. This distinction avoids treating a plausible timing hypothesis as ground truth.

Algorithm 1 Conservative two-edge event timing audit.

Require: 8-s aggregate x_t, appliance powers $P_{t,i}$, states $z_{t,i}$, thresholds τ_i, and $L = 2$ samples.
Ensure: Audited appliance series $\widetilde{P}_{t,i}$ and states $\widetilde{z}_{t,i}$; aggregate x_t is unchanged.

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Set P̃_{t,i} ← P_{t,i} and z̃_{t,i} ← z_{t,i}.
for each target appliance i ∈ {1, ..., M} do
    for each complete on period [s, e) in z_{t,i} do
        Reject the event if an endpoint is unavailable, either appliance edge is below max(40, 0.3τ_i), or another monitored target has a transition of at least 40 W within L samples of either boundary.
        Form the admissible lag set using (??); discard candidates whose two relative edge errors exceed one.
        if C_i(ℓ*) ≤ 0.75, the score margin is at least 0.20, and the shifted interval does not overlap another ON interval then
            Move the excess profile and the complete state interval by ℓ* using (??).
        end if
    end for
end for
Verify Σ_t P̃_{t,i} = Σ_t P_{t,i} for every appliance and x_t is bitwise unchanged.
return x_t, P̃_{t,i}, and z̃_{t,i}.
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